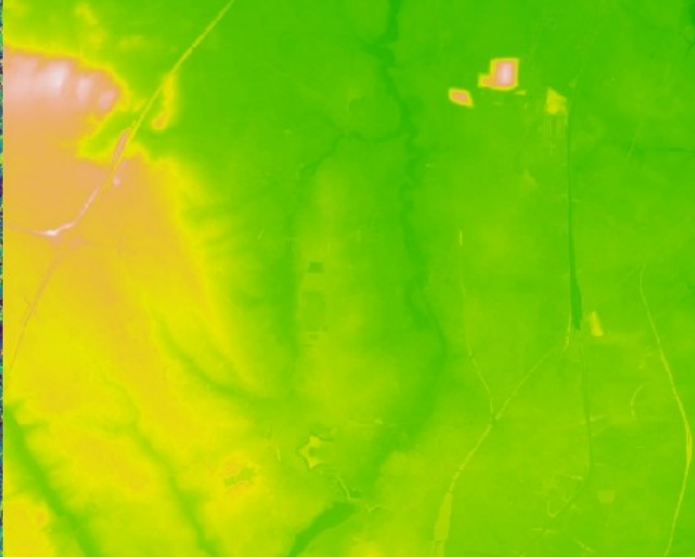
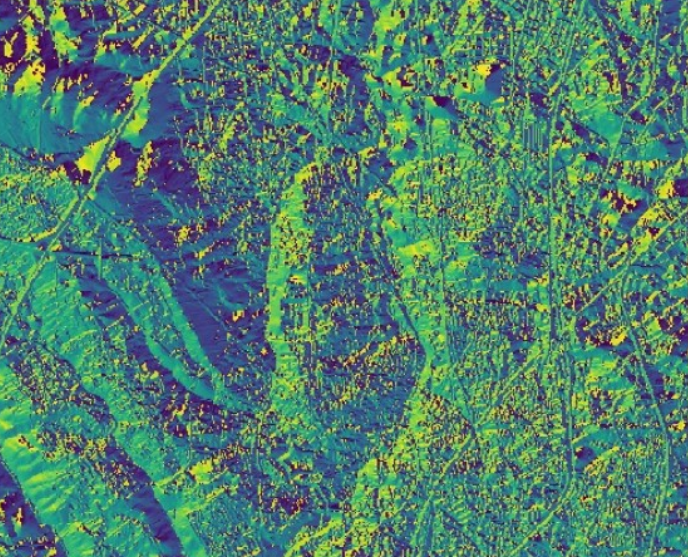
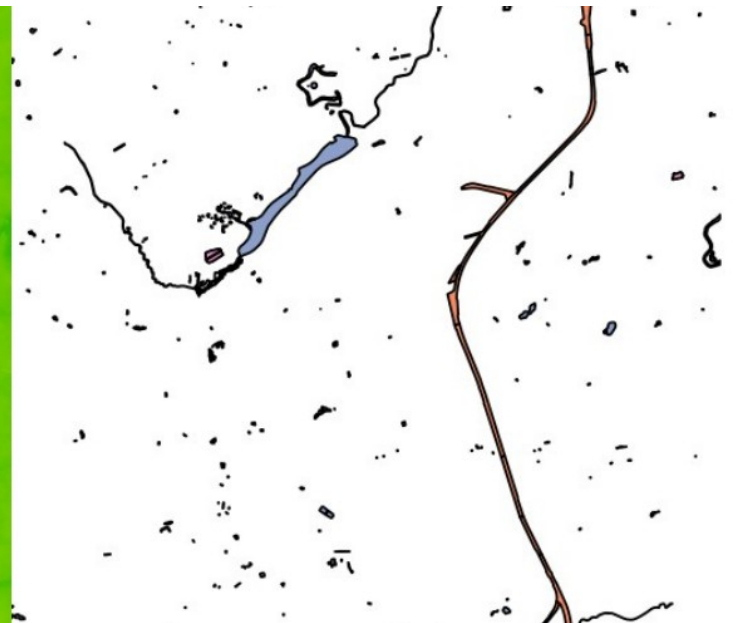
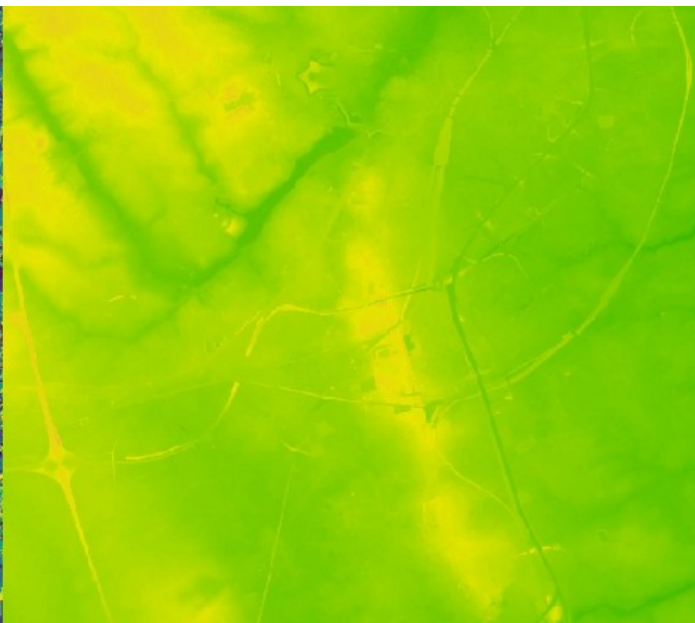
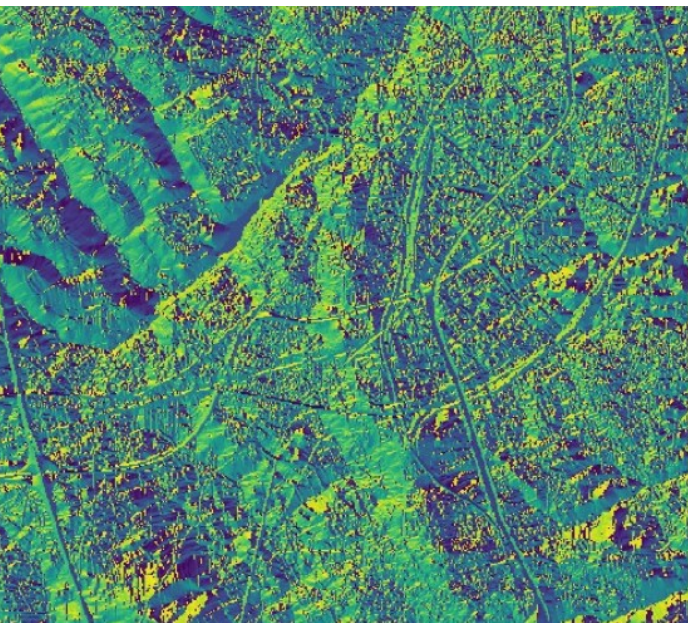




GIS in der Landschaftsökologie

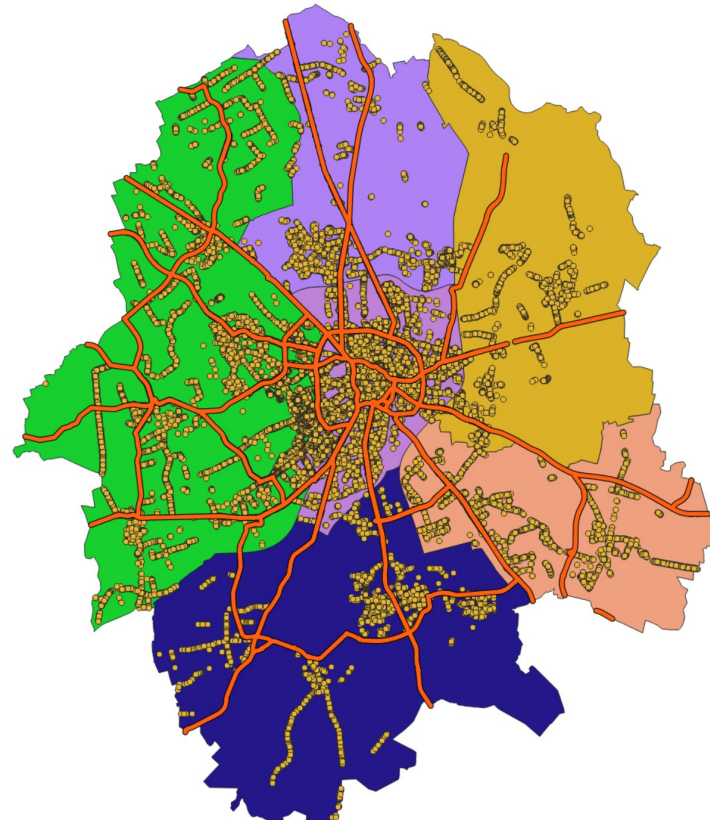
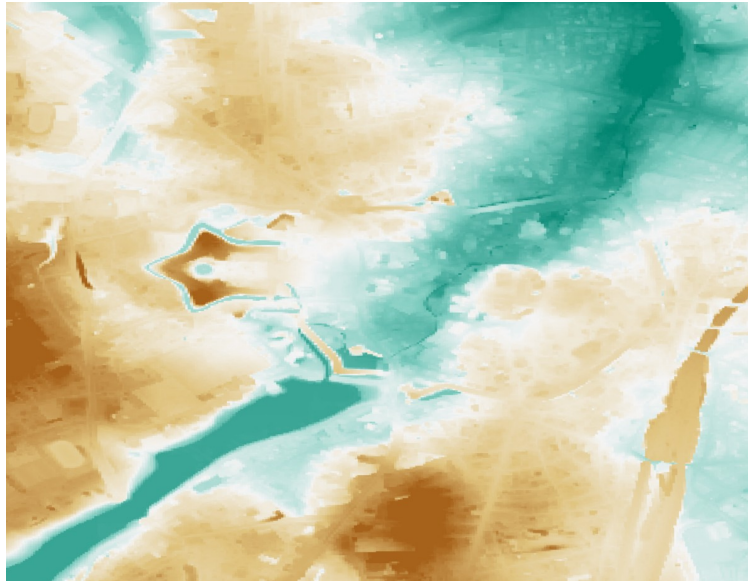
Hanna Meyer & Marvin Ludwig

Referenzsysteme & Projektionen



Besonderheiten geographischer Daten

- **Koordinaten stellen Raumbezug her**
- Beschreibung und Analyse der Beziehung zwischen Positionen / geographische Entitäten / Objekte
- In Kombination mit Eigenschaften der Position / des Objekts
- Modell der “realen Umwelt”



Woher kommen Koordinaten?

Referenzsysteme in QGIS

Coordinate Reference System	Authority ID
EPSG:3035 - ETRS89-extended / LAEA Europe	EPSG:3035
EPSG:32732 - WGS 84 / UTM zone 32S	EPSG:32732
EPSG:32641 - WGS 84 / UTM zone 41N	EPSG:32641
EPSG:32639 - WGS 84 / UTM zone 39N	EPSG:32639
EPSG:32632 - WGS 84 / UTM zone 32N	EPSG:32632

Predefined Coordinate Reference Systems

☐ Hide deprecated CRSs

Coordinate Reference System	Authority ID
ETRS89 / UTM zone 29N (N-E)	EPSG:3041
ETRS89 / UTM zone 30N	EPSG:25830
ETRS89 / UTM zone 30N (N-E)	EPSG:3042
ETRS89 / UTM zone 31N	EPSG:25831
ETRS89 / UTM zone 31N (N-E)	EPSG:3043
ETRS89 / UTM zone 32N	EPSG:25832
ETRS89 / UTM zone 32N (N-E)	EPSG:3044
ETRS89 / UTM zone 33N	EPSG:25833
ETRS89 / UTM zone 33N (N-E)	EPSG:3045
ETRS89 / UTM zone 34N	EPSG:25834
ETRS89 / UTM zone 34N (N-E)	EPSG:3046

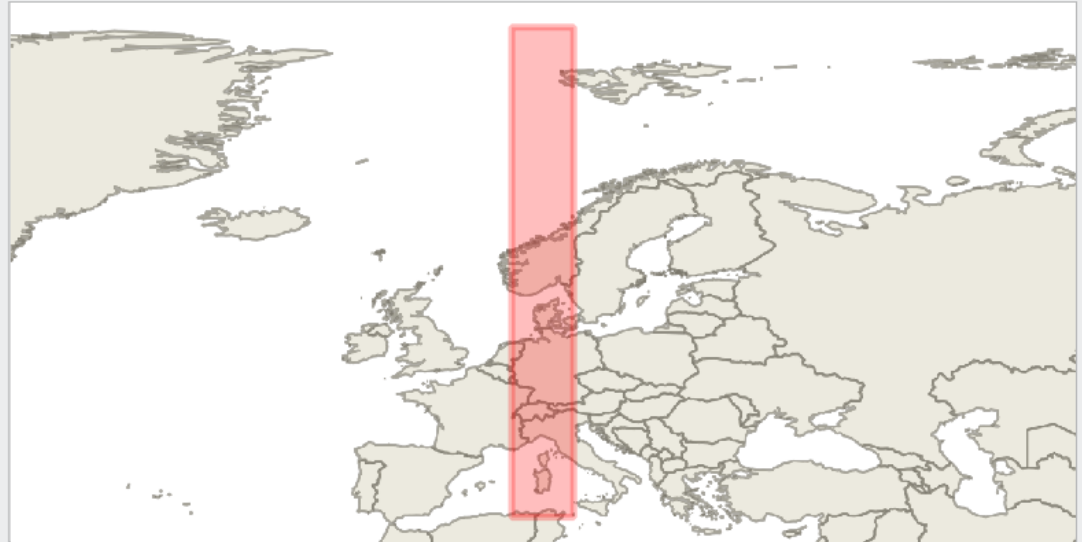
ETRS89 / UTM zone 32N

Properties

- Units: meters
- Dynamic (relies on a datum which is not plate-fixed)
- Celestial body: Earth
- Based on *European Terrestrial Reference System 1989 ensemble* (EPSG:6258), which has a limited accuracy of **at best 0.1 meters**.
- Method: Universal Transverse Mercator (UTM)

WKT

```
PROJCRS["ETRS89 / UTM zone 32N",
  BASEGEOGCRS["ETRS89",
    ENSEMBLE["European Terrestrial Reference System 1989 ensemble",
      MEMBER["European Terrestrial Reference Frame 1989"],
      MEMBER["European Terrestrial Reference Frame 1990"],
      MEMBER["European Terrestrial Reference Frame 1991"]
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Referenzsysteme in QGIS

Coordinate Reference System	Authority ID
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EPSG:32732 - WGS 84 / UTM zone 32S	EPSG:32732
EPSG:32641 - WGS 84 / UTM zone 41N	EPSG:32641
EPSG:32639 - WGS 84 / UTM zone 39N	EPSG:32639
EPSG:32632 - WGS 84 / UTM zone 32N	EPSG:32632

LAEA

WGS84

Predefined Coordinate Reference Systems

☐ Hide deprecated CRSs

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ETRS89 / UTM zone 32N (N-E)	EPSG:3044
ETRS89 / UTM zone 33N	EPSG:25833
ETRS89 / UTM zone 33N (N-E)	EPSG:3045
ETRS89 / UTM zone 34N	EPSG:25834
ETRS89 / UTM zone 34N (N-E)	EPSG:3046

UTM

ETRS89

EPSG

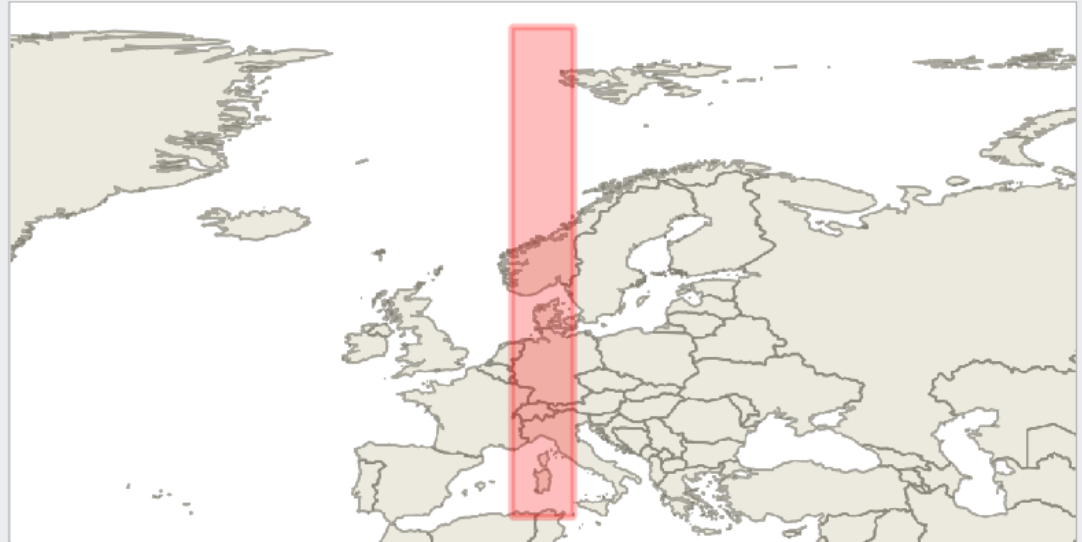
ETRS89 / UTM zone 32N

Properties

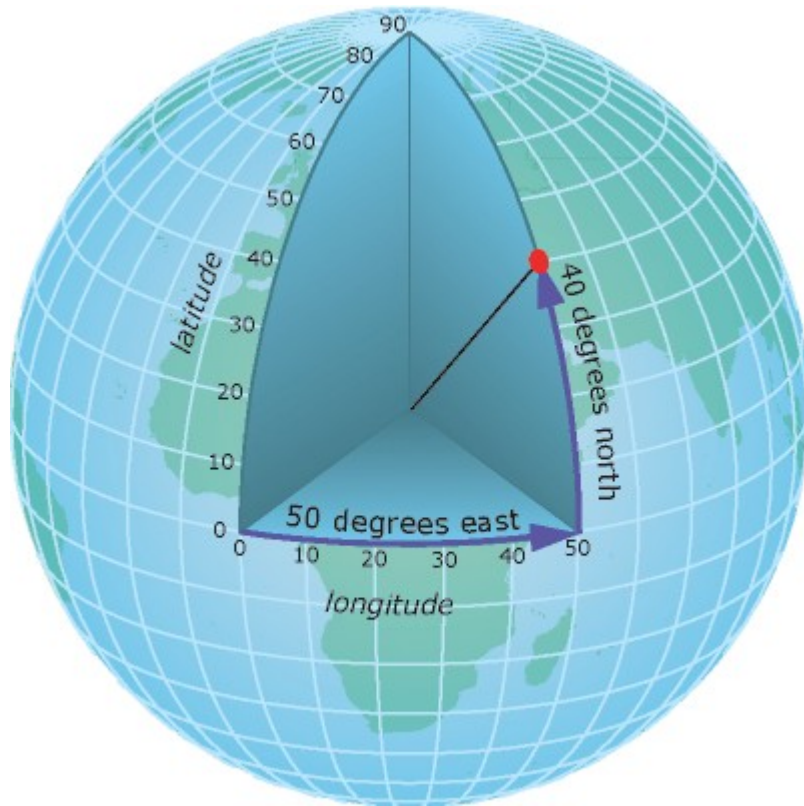
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      MEMBER["European Terrestrial Reference Frame 1990"],
      MEMBER["European Terrestrial Reference Frame 1991"]
```



WGS84: World Geodetic System 1984



Münster in Deutschland

KOORDINATEN

Breitengrad	51.9607
Längengrad	7.6261
Breite in Grad	51° 57' 39" N
Länge in Grad	7° 37' 34" E
Höhe über NN	66 m

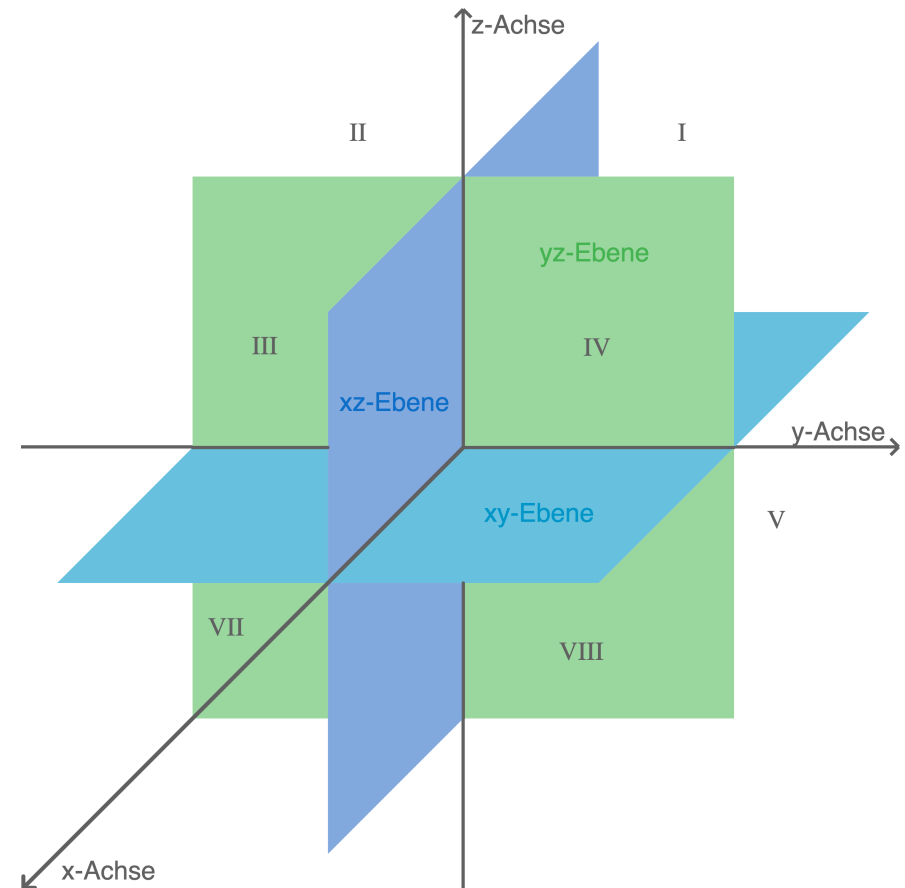
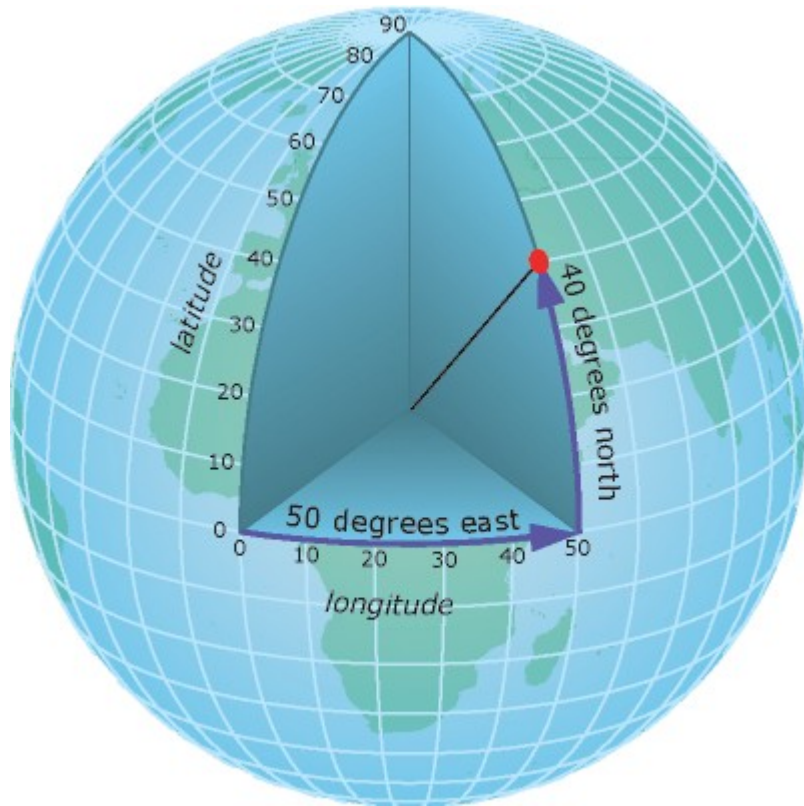


Rio de Janeiro in Brasilien

KOORDINATEN

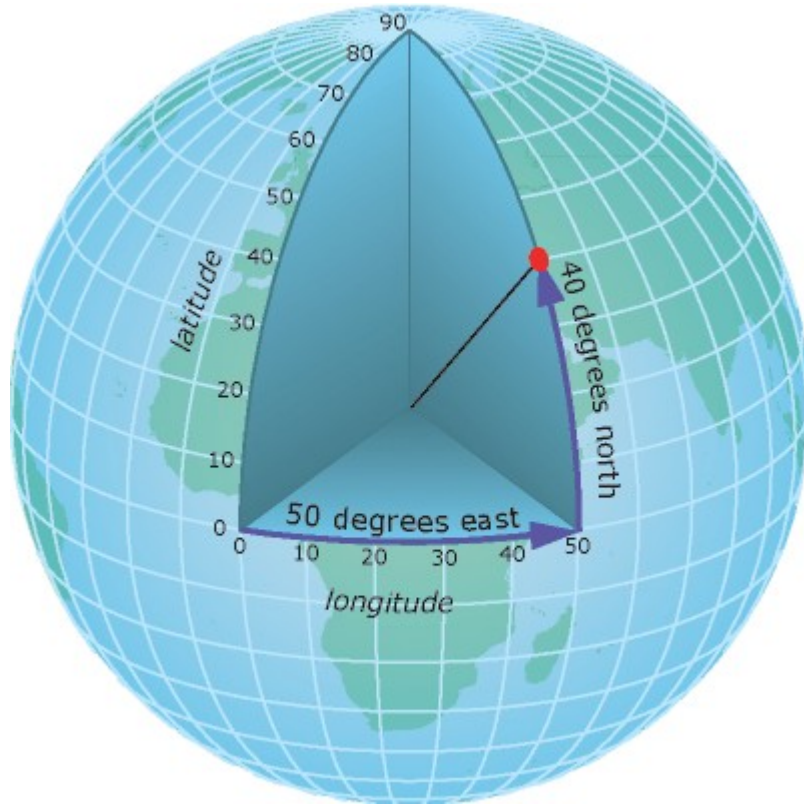
Breitengrad	-22.9083
Längengrad	-43.1964
Breite in Grad	22° 54' 30" S
Länge in Grad	43° 11' 47" W
Höhe über NN	6 m

Geographisch vs. Kartesisch



Was sind Vor- und Nachteile der beiden Systeme?

WGS84: World Geodetic System 1984



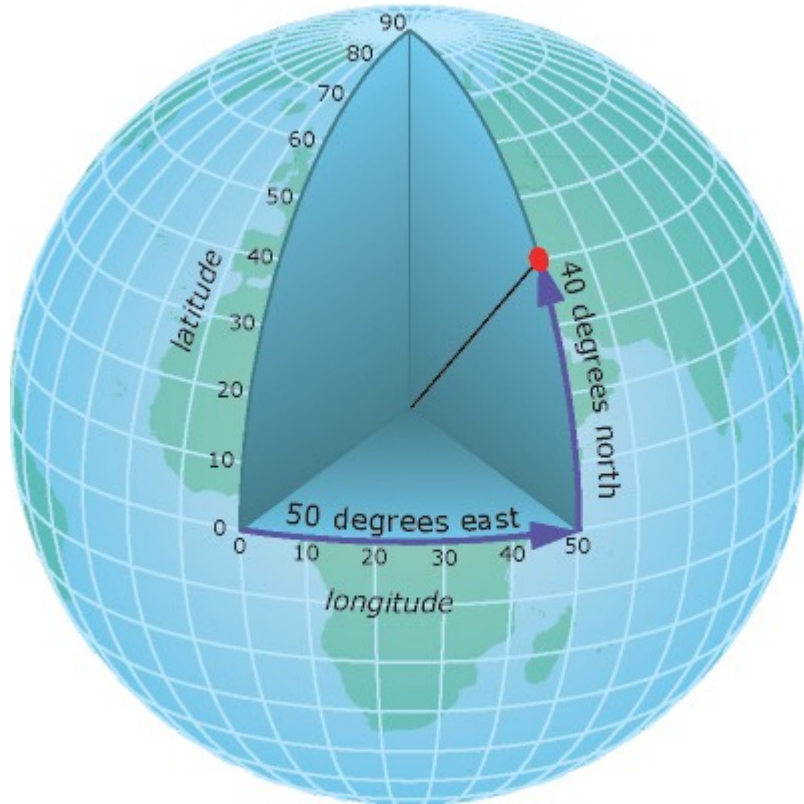
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Breite in Grad	51° 57' 39" N
Länge in Grad	7° 37' 34" E
Höhe über NN	66 m

- Die Erde ist keine Kugel
- Koordinaten richten sich nach sgn. Referenzellipsoid
- WGS84 ist die Bezeichnung des Referenzellipsoides

ETRS89 Europäisches Terrestrisches Referenzsystem 1989



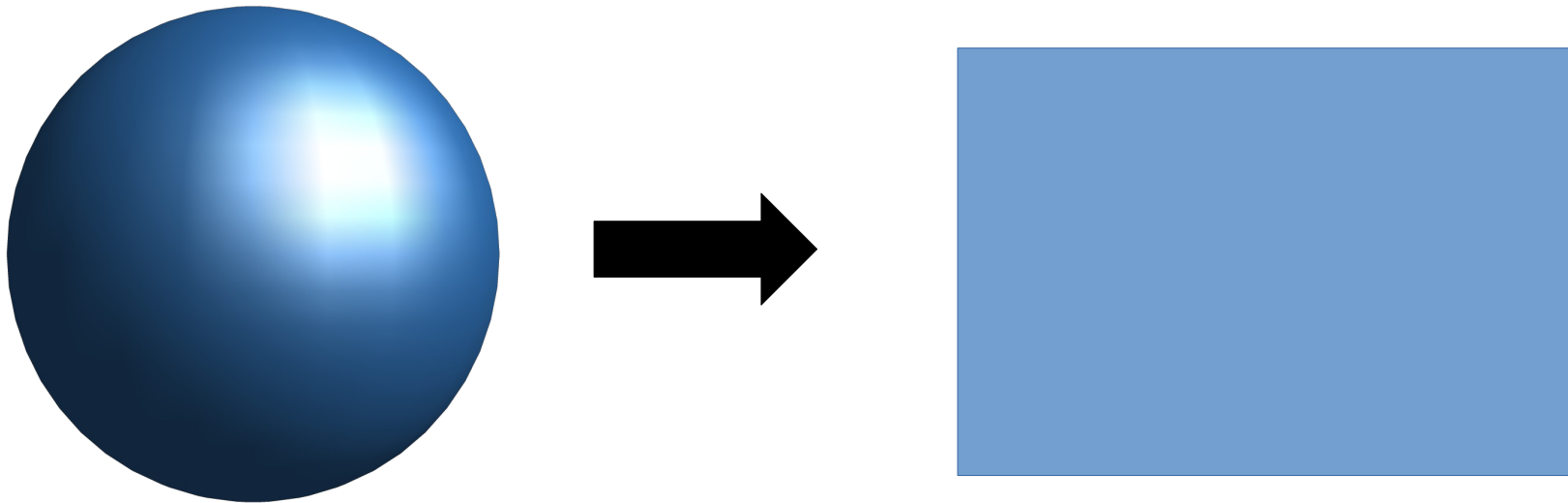
- Verwendet leicht anderen Referenzellipsoid mit dem Namen GRS80
- Unterschiede zwischen WGS84 und ETRS89 ~ 1m

Projektionen

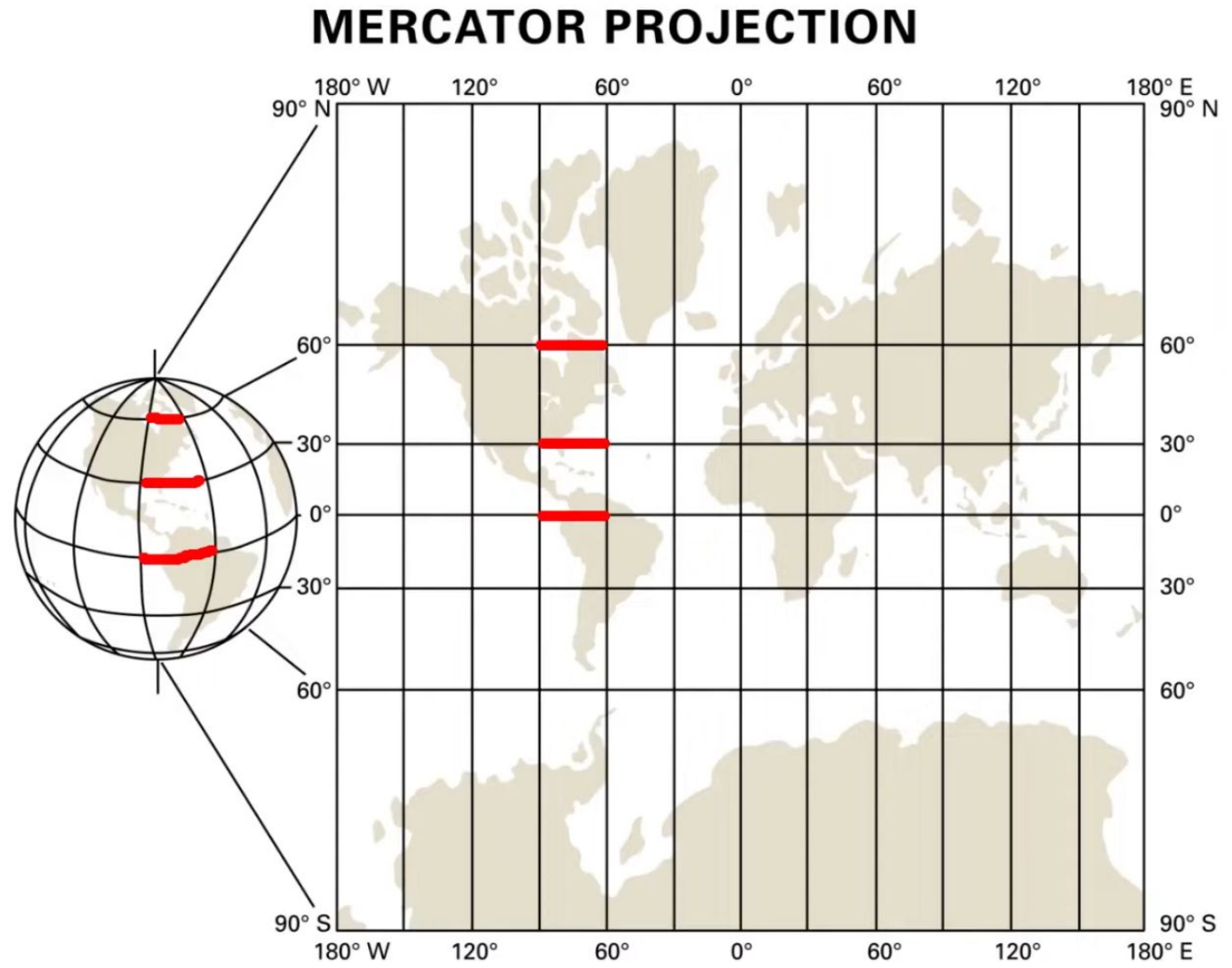
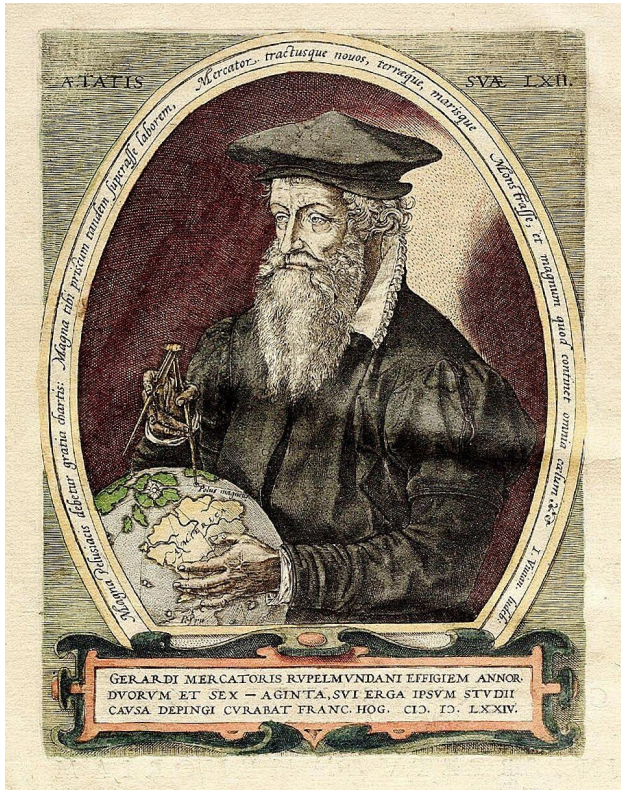


Altes Gaskraftwerk Meppen © IMAGO/Zoonar.com/flight-pictures

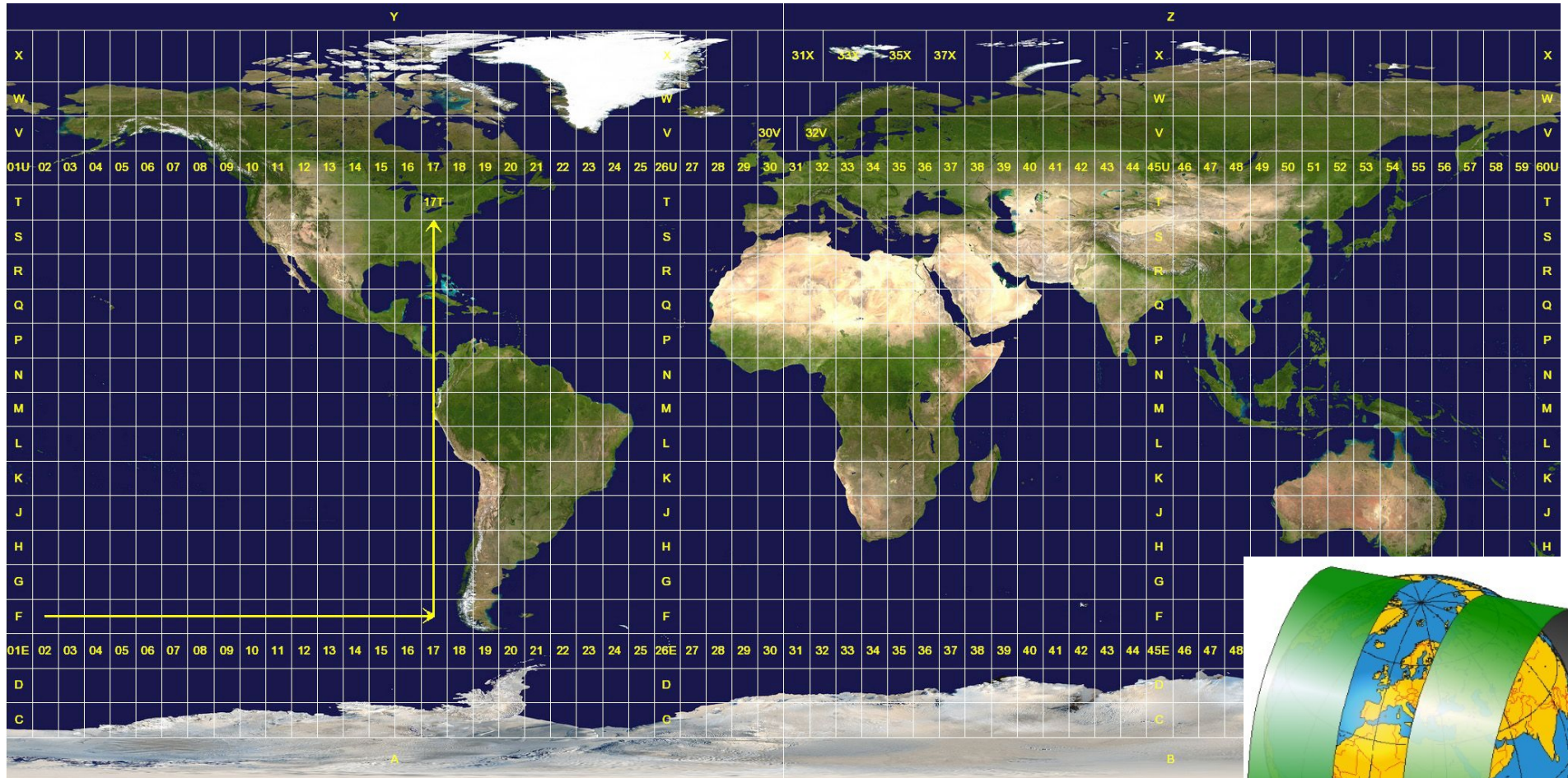
Projektionen



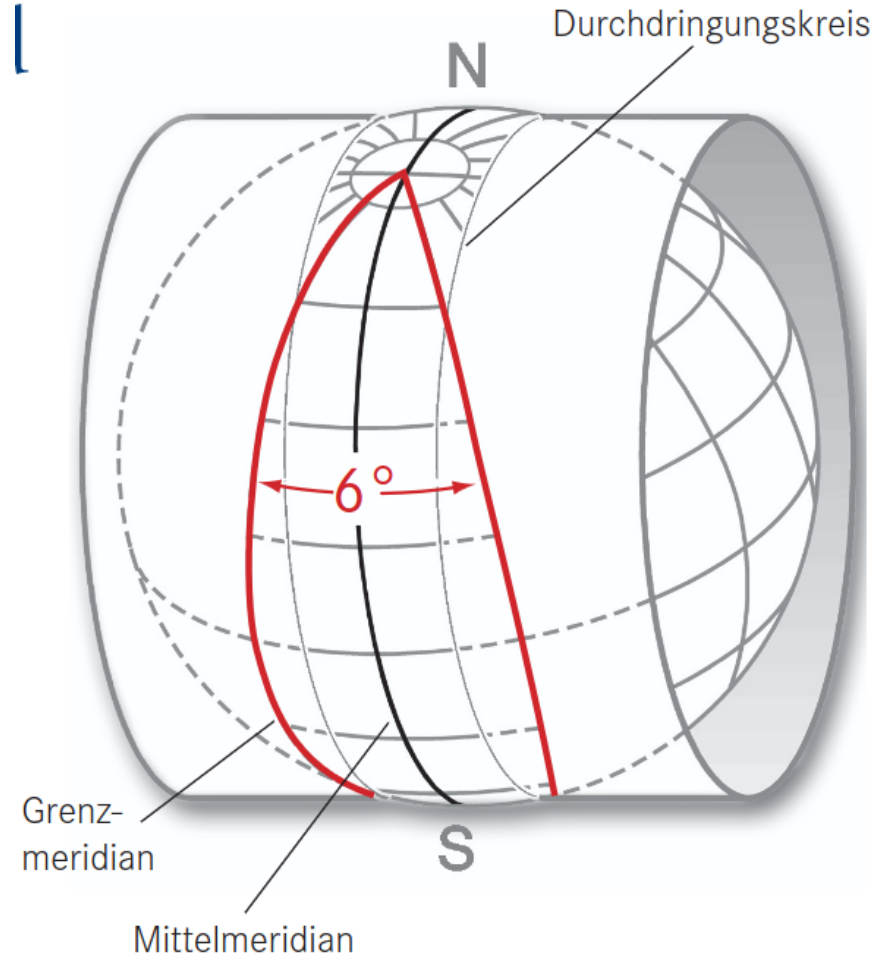
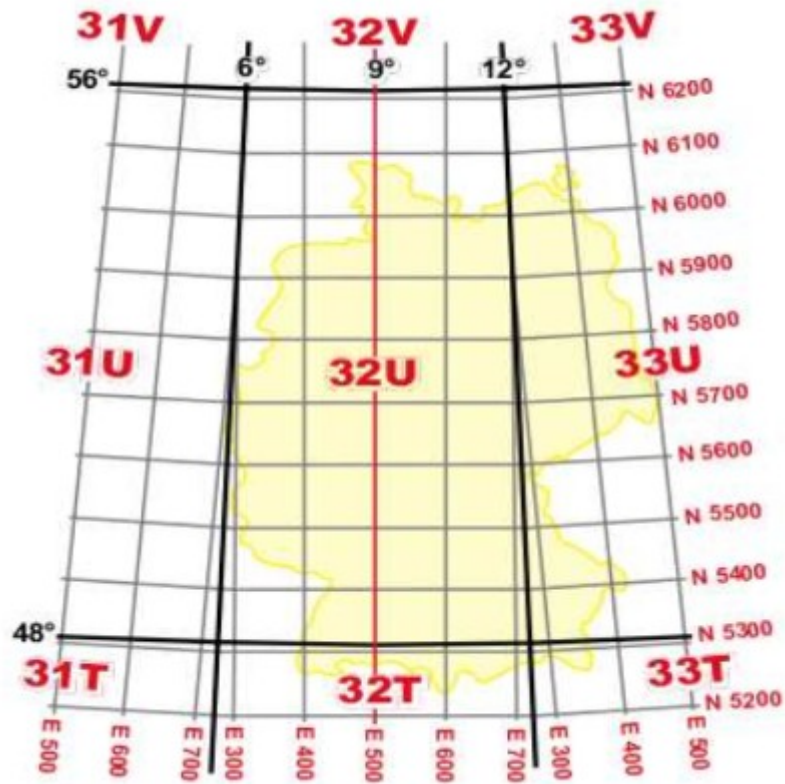
Die Mercator Projektion



Universal Transverse Mercator



UTM in Deutschland



Schematische Darstellung der querachsigen Schnitzylinder-Abbildung mit Meridianstreifen von 6° Ausdehnung.

Labert Azimutal

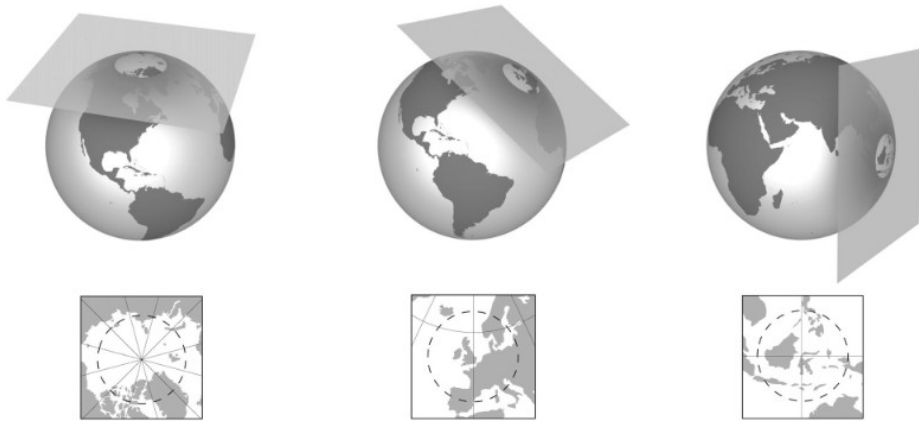


Johann Heinrich Lambert
(Lithographie von Godefroy Engelmann, 1829)

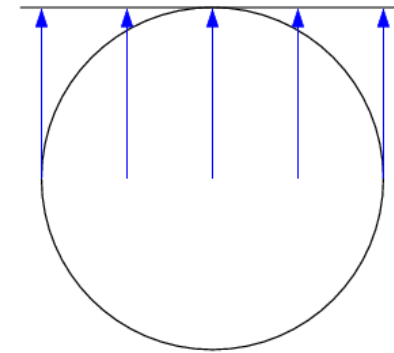


ETRS89-LAEA

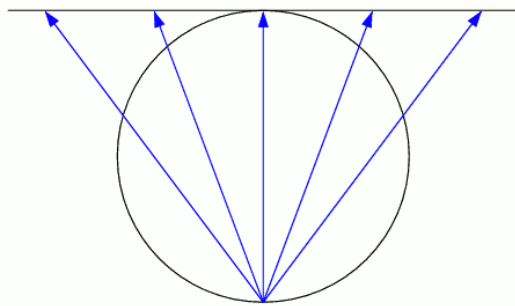
Azimutale Projektionen



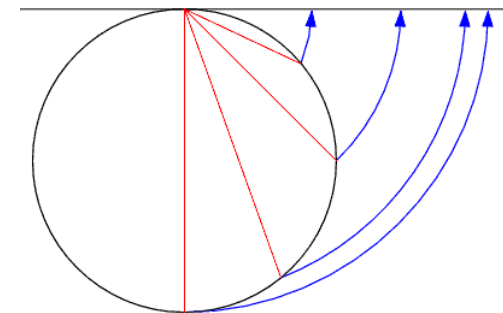
Savric et al. 2016



Orthografisch



Stereografisch



Lambert Azimutal Equal Area

Referenzsysteme in QGIS

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EPSG:32641 - WGS 84 / UTM zone 41N	EPSG:32641
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EPSG:32632 - WGS 84 / UTM zone 32N	EPSG:32632

LAEA

WGS84

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ETRS89 / UTM zone 31N (N-E)	EPSG:3043
ETRS89 / UTM zone 32N	EPSG:25832
ETRS89 / UTM zone 32N (N-E)	EPSG:3044
ETRS89 / UTM zone 33N	EPSG:25833
ETRS89 / UTM zone 33N (N-E)	EPSG:3045
ETRS89 / UTM zone 34N	EPSG:25834
ETRS89 / UTM zone 34N (N-E)	EPSG:3046

UTM

ETRS89

EPSG

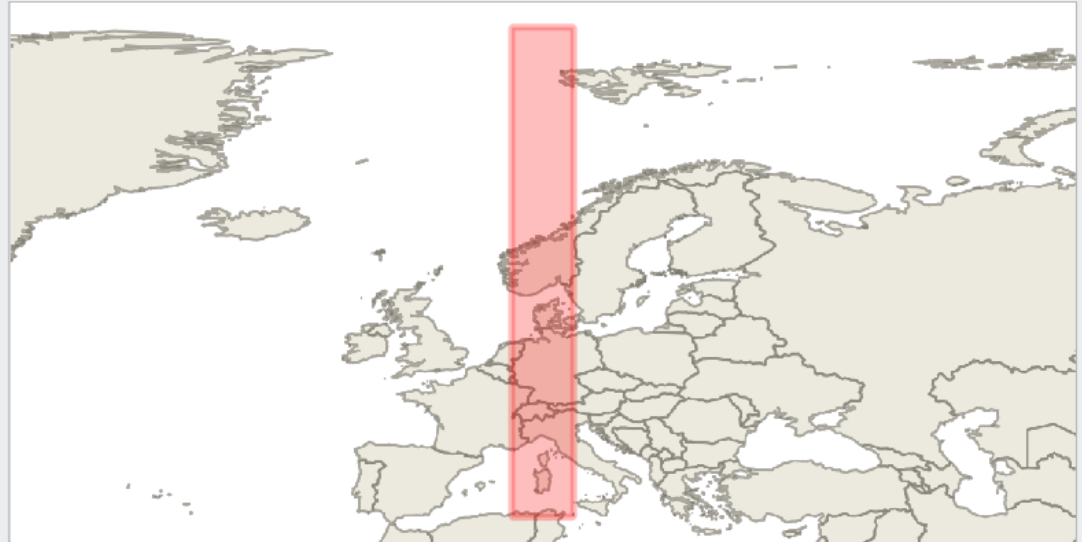
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  BASEGEOGCRS["ETRS89",
    ENSEMBLE["European Terrestrial Reference System 1989 ensemble",
      MEMBER["European Terrestrial Reference Frame 1989"],
      MEMBER["European Terrestrial Reference Frame 1990"],
      MEMBER["European Terrestrial Reference Frame 1991"]]
```



Referenzsysteme in GIS

- Geodaten werden in einem bestimmten Referenzsystem erstellt und verbreitet
 - European Environmental Agency: ETRS89-LAEA
 - OpenStreetMap: WGS84
 - Sentinel-2: WGS84/UTM
 - Baumkatalog der Stadt Münster: ETRS89-UTM52N
- Im GIS System werden Daten mitunter dargestellt als wären sie in der gleichen Projektion. **Das ist rein visuell!**
- Bei der Verarbeitung von Geodaten müssen diese im selben Projektionssystem sein.

Realbeispiel: LIFECrossBorderBog



Onze Lieve Vrouwetoren in Amersfoort

X_COORDINAAT (RD)	262738
Y_COORDINAAT (RD)	468289

X Koordinate (UTM)	360023
Y Koordinate (UTM)	5783027

